

Daniel Niechwiadowicz

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EDUCATION

University of Houston

Houston, TX

Bachelor of Science in Computer Science, GPA: 3.47

Expected May 2027

- **Coursework:** Fundamentals of AI, Database Systems, Data Structures, Data Algorithms, Operating Systems, Computer Networking, Discrete Mathematics

PROJECTS

CNN Clothing Image Classification | *Python, PyTorch, CUDA, scikit-learn*

Spring 2026

- Designed and trained 6 convolutional neural networks from scratch in PyTorch (85K–1.8M parameters) to classify 1,000 real-world clothing images into 10 categories
- Built a preprocessing pipeline and custom PyTorch Dataset, then ran controlled experiments on depth, width, kernel size, and augmentation; deeper networks improved accuracy by 4–5 points
- Evaluated models with per-class precision, recall, weighted F1, and confusion matrices; applied BatchNorm, dropout, early stopping, and learning-rate scheduling

Texas Hold'em Poker AI | *Python, NumPy, Reinforcement Learning*

Spring 2026

- Built a reinforcement learning agent trained through self-play, structured as a modular Python package (game engine, state encoder, policy network, evaluator)
- Engineered a 318-dimensional state encoder that converts live game states into normalized float32 feature vectors covering cards, pot and stack ratios, position, and betting history (aggression ratios, pot odds)

Predictive Modeling: Regression & Classification | *Python, scikit-learn, Pandas, NumPy*

Fall 2026

- Built multivariable linear regression models on 20,640 California housing records with a 70/15/15 train/validation/test split, cutting validation RMSE by 34% versus a mean-prediction baseline
- Compared logistic regression, Gaussian Naive Bayes, and KNN ($k = 1, 5, 15$) for malignancy prediction using stratified splits, feature standardization, precision, recall, F1, and ROC-AUC

Traffic Stop Patterns Analysis | *Python, Pandas, Matplotlib*

Spring 2026

- Built an end-to-end ETL pipeline (ingestion, cleaning, aggregation, output) to clean and standardize a large real-world traffic stop dataset
- Developed Pandas aggregation pipelines to identify enforcement trends across time and districts, and visualized peak enforcement hours and speeding concentrations

Post Office 8: Full-Stack Database Application | *MySQL, SQL, Node.js/Express, React*

Spring 2026

- Designed a REST API on MySQL with a transactional insert flow across 6 tables and full rollback on failure, plus JWT authentication and role-based access control
- Wrote SQL queries including self-joins and seeded a 72-row pricing table across weight bands, 9 shipping zones, and 3 package types; debugged schema mismatches in production

EXPERIENCE & LEADERSHIP

Team Leader

2025 – 2026

Chick-fil-A

Katy, TX

- Lead time-sensitive, high-volume operations and train new team members on structured procedures

Co-Leader

2019 – 2024

VEX Robotics Team, Sushi Warriors

- Designed and documented robot control algorithms and led competition strategy; qualified for the VEX World Championship 4 of 5 years and won 4 state titles

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

ML & Data: PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, CUDA

Methods: Deep Learning, CNNs, Reinforcement Learning, Regression, Classification, ETL, Feature Engineering

Tools: MySQL, Jupyter Notebook, Node.js/Express, REST APIs, Git/GitHub